Graph limits for shortest directed paths in bipolar-oriented triangulations

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Abstract

We consider shortest directed-path distances between edges of the uniform infinite bipolar-oriented triangulation. Each edge is represented by its two primal Peano times, and distances are normalized by a median of the maximal finite directed distance in a KMSW cell. We prove that every joint subsequential limit of the domains and the distance graphs, in the Fell topology, is almost surely the graph of a locally bounded continuous function. The proof controls shortest-path costs on finite subtrees of the four extremal trees. Its probabilistic starting point is an exponential upper tail for the directed diameter of a boundary segment in a half-map. We also obtain exponential upper tails for the maximal finite SDP distance in an ordinary KMSW cell and show that its medians are comparable to $n^{3/8}$. The proof treats pairs near the boundary of the reachability domain and does not assume regular variation of the median sequence. In particular, the transverse-label argument is stated directly in terms of the GHS north and south pockets and their exact conditional laws, and the boundary-cap argument uses only the stated KMSW-cell crossing estimate of Borga–Gwynne.

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1 Statement and conventions

Let M be the uniform infinite bipolar-oriented triangulation (UIBOT), with its directed root edge. We use the KMSW encoding convention

$$Z = (L, R) : \mathbb{Z} \to \mathbb{Z}^2, \qquad Z(0) = 0, \qquad Z(t) - Z(t-1) \text{ uniform on } \{X, L_0, R_0\}, \tag{1.1}$$

independently for different t, where

$$X = (1, -1), \qquad L_0 = (-1, 0), \qquad R_0 = (0, 1).$$

The symbols L_0, R_0 denote letters, whereas L, R denote coordinates of the walk.

Let

$$\lambda^N, \lambda^W : \mathbb{Z} \to E(M)$$

be the two primal Peano orders, both indexed so that time zero is the root edge. The N-order is the ordinary KMSW order. The W-order is the order obtained by reflecting the KMSW exploration across the orientation axis, equivalently the primal west-going Peano order of [5, Remark 1.8]. We choose its sign so that both orders increase along directed paths. The primal W-encoding has the same i.i.d. increment law as the N-encoding; the dual west-going encoding need not have that increment law. Write $Z^N = Z$ and Z^W for the two primal encoding walks. The equality of the primal and dual west-going edge orders is [5, Lemma 2.5].

For an edge e, write

$$i(e) = (\lambda^N)^{-1}(e), \qquad j(e) = (\lambda^W)^{-1}(e), \qquad p(e) = (i(e), j(e)). \tag{1.2}$$

Write $e \preceq f$ when a directed path can have e as its first edge and f as its last edge, including $e = f$. We will prove the two-order characterization

$$e \preceq f \quad \Longleftrightarrow \quad p(e) \le p(f) \text{ coordinatewise.} \tag{1.3}$$

Let $d^\bullet_M(e, f)$ be the least number of edges in such a path, with $d^\bullet_M(e, e) = 1$. For integers $a \le b$, the cell $M_{a,b}$ consists of the active edges with N-labels in $[a, b] \cap \mathbb{Z}$, with their inherited endpoint identifications; missing boundary edges are not included. Put

$$\Delta_n = \max\{\mathrm{SDP}_{M_{0,n}}(u, v) : u, v \in V(M_{0,n}),\ \mathrm{SDP}_{M_{0,n}}(u, v) < \infty\}, \qquad n \ge 0. \tag{1.4}$$

Thus $M_{0,n}$ has $n+1$ active edges and $1 \le \Delta_n \le n+1$. For each $n \ge 1$, fix any median m_n of Δ_n:

$$\mathbb{P}[\Delta_n \le m_n] \ge \frac{1}{2}, \qquad \mathbb{P}[\Delta_n \ge m_n] \ge \frac{1}{2}.$$

Define the closed discrete sets

$$U_n = \{n^{-1}p(e) : e \in E(M)\} \subset \mathbb{R}^2, \qquad W_n = \{(x,y) \in U_n^2 : x \le y\} \subset \mathbb{R}^4, \tag{1.5}$$

and

$$D_n(n^{-1}p(e), n^{-1}p(f)) = m_n^{-1} d_M^{\bullet}(e,f), \quad e \preceq f, \qquad \Gamma_n = \mathrm{Graph}(W_n, D_n) \subset \mathbb{R}^5. \tag{1.6}$$

A compact base box contains only finitely many points of W_n.

Let $\mathsf{F}(\mathbb{R}^d)$ denote the space of closed subsets of $\mathbb{R}^d$, including the empty set, with the Fell topology. Equivalently, this is the Hausdorff topology after adjoining the point at infinity in a fixed one-point compactification metric.

Theorem 1.1 (Graph limits)**.** *Every subsequential weak limit of*

$$(W_n, \Gamma_n) \quad \text{in } \mathsf{F}(\mathbb{R}^4) \times \mathsf{F}(\mathbb{R}^5)$$

is almost surely of the form

$$(W, \mathrm{Graph}(W, D)),$$

where $D : W \to [0,\infty)$ is a unique locally bounded continuous function. In particular, every fiber of the limiting closed set over its limiting domain is a singleton, simultaneously for all points of that domain.

The uniqueness is uniqueness of the function represented by a given limiting graph, not uniqueness of the subsequential law.

Theorem 1.2 (Cell diameter and normalization)**.** *There are deterministic constants $c, C \in (0,\infty)$ such that*

$$\mathbb{P}[\Delta_n > A(n+1)^{3/8}] \le Ce^{-cA}, \qquad n \ge 0,\ A \ge 1. \tag{1.7}$$

For every choice of medians in (1.6),

$$cn^{3/8} \le m_n \le Cn^{3/8}, \qquad n \ge 1. \tag{1.8}$$

Throughout, constants denoted by c, C can change between occurrences. They are deterministic and independent of the location and length of a deterministic clock interval. Write

$$\sigma = \frac{3}{8}, \qquad \mathrm{osc}_I\, Z = \max\Big\{ \max_I L - \min_I L,\ \max_I R - \min_I R \Big\}.$$

Distance convention. Subdivide every map edge at a black midpoint and give each half-edge length $1/2$. The original vertices are called white. We use $d = d_M$ for the directed shortest-path distance on this subdivided graph, allowing $+\infty$ for unreachable pairs. For comparable black points,

$$d(e,f) = d_M^{\bullet}(e,f) - 1. \tag{1.9}$$

Thus $d(e,e) = 0$, and all triangle inequalities below are exact for d. By (1.8), restoring the original convention changes normalized heights by $m_n^{-1} \to 0$.

Structure of the proof. We first prove a boundary-segment tail in a half-map. A dyadic decomposition converts this into uniform estimates for finite extremal-tree subtrees. The GHS north/south pocket law then controls all transverse labels of one ordinary cell, yielding Theorem 1.2. Finally, the tree estimates give the endpoint modulus needed for Theorem 1.1.

2 Locality and deterministic planar comparisons

2.1 KMSW endpoint locality

For a finite KMSW word, write $\lambda_t = (u_t, v_t)$ for the active edge. The tail frontier is a stack on which X pushes, L_0 pops, and R_0 holds. The head frontier is a stack on which R_0 pushes, X pops, and L_0 holds. When a stack underflows, a fresh parent boundary vertex is exposed. The only transfer of a vertex from the head frontier to the tail frontier occurs at an X step; these are the triangular sewing rules of [6, Section 2.2], see also [2, Section 3.1].

Lemma 2.1 (Signed endpoints and interval locality)**.** *For $i \le j$,*

$$u_i = u_j \iff L(i) = L(j) = \min_{i \le t \le j} L(t), \tag{2.1}$$

$$v_i = v_j \iff R(i) = R(j) = \min_{i \le t \le j} R(t). \tag{2.2}$$

For $i < j$, $v_i = u_j$ if and only if there is an X step at some $q \in (i, j] \cap \mathbb{Z}$ with

$$R(i) = R(q-1) = \min_{i \le t \le q-1} R(t), \qquad L(q) = L(j) = \min_{q \le t \le j} L(t). \tag{2.3}$$

For $i \ge j$, one has $v_i \ne u_j$. Consequently active labels strictly increase along every directed path. The edge-spanned directed multigraph on labels $[a, b] \cap \mathbb{Z}$ is determined, with all signed endpoint incidences, by the increments in $(a, b] \cap \mathbb{Z}$ and is isomorphic to the fresh sewing of that word.

Proof. For a stack height H, the states at times $i \le j$ coincide exactly when $H(i) = H(j) = \min_{[i,j]} H$. Descending below that level leaves the state permanently, while every push above the level must be popped before the same state is recovered. This remains valid at a newly exposed parent, proving (2.1)–(2.2).

A vertex has all of its head occurrences before all of its tail occurrences. An R_0 step creates a fresh head, an L_0 step exposes the parent of the current tail, and an X step transfers the old head to the tail stack. No move identifies two previously distinct vertices or transfers a tail state back to the head stack. Hence a common head and later tail must pass through one X step, which gives (2.3); the same observation excludes $v_i = u_j$ for $i \ge j$.

All signed endpoint equalities between edges labeled in $[a, b]$ are therefore determined by the increments in $(a, b]$. Subtracting $Z(a)$ changes none of the tests. These equalities specify the edge-labeled directed multigraph, proving the last assertion. □

The final assertion concerns the graph of active edges. It does not assert that missing edges or planar completions of a fresh interval agree with those supplied by its surroundings.

Lemma 2.2 (Two-buffer bound and cell recursion)**.** *If v is a consecutive subword of a finite word w, then*

$$\Delta(w) \ge \Delta(v) - 2. \tag{2.4}$$

This remains true after arbitrary resampling outside v. For a uniform finite word and integers $k, h \geq 1$,

$$\mathbb{P}[\Delta_n \geq k(h+2)] \leq \mathbb{P}[\Delta_n \geq h]^k. \tag{2.5}$$

Proof. Take a geodesic $e_1 \cdots e_q$ in the fresh subword, with $q = \Delta(v) \geq 3$. A shortcut in a completion between the endpoint adjacent to e_1 and the endpoint adjacent to e_q would, after adjoining e_1 and e_q, use only active labels strictly between the two anchor labels. By Lemma 2.1, the same shortcut would already exist in the fresh subword. The middle distance therefore remains at least $q-2$ in every completion, proving (2.4).

On $\{\Delta_n \geq k(h+2)\}$ choose a geodesic and divide its first $k(h+2)$ edges into k consecutive packets of $h+2$ edges. If a_ℓ, b_ℓ are the first and last active labels in packet ℓ, the intervals $(a_\ell, b_\ell]$ are pairwise disjoint because active labels strictly increase along a directed path. Each packet is a geodesic in its fresh interval: an anchored shortcut would be local by Lemma 2.1. Fixing the increments in $(a_\ell, b_\ell]$ therefore forces the event $\{\Delta_n \geq h\}$ regardless of all increments outside that interval, by (2.4). Thus $\{\Delta_n \geq k(h+2)\}$ is contained in the k-fold disjoint occurrence of $\{\Delta_n \geq h\}$. Reimer's inequality for arbitrary events in a finite product space gives (2.5); see [7, 4]. □

2.2 The four extremal trees

In an upward drawing, define

$$\mathcal{T} = \{T_R^+, T_L^+, T_R^-, T_L^-\}.$$

In T_R^+, the parent of a white vertex is the black point on its rightmost outgoing edge, and the parent of a black point is the head of that edge. Define T_L^+ using leftmost outgoing edges. In T_R^-, the parent of a white vertex is the black point on its rightmost incoming edge and the parent of a black point is its tail; define T_L^- similarly. Thus the $+$ trees are rooted toward the future and the $-$ trees toward the past. A directed segment in a $-$ tree runs away from its root. Every black point belongs to all four subdivided trees.

In the UIBOT, two branches of each of these trees coalesce almost surely, by the stack/contour description and recurrence of the corresponding lazy coordinate walk. The clock associated with each tree is

$$c(T_R^+) = c(T_L^-) = N, \qquad c(T_L^+) = c(T_R^-) = W. \tag{2.6}$$

With our convention, the first pair is the ordinary KMSW cut-tree pair and the second is its reflected counterpart. Reversing orientations or reflecting the map permutes these four rooted configurations. At a deterministic time in the associated clock, the corresponding half-map has the ordinary i.i.d. sewing law.

For points a, a' in a tree T, let $q_T(a, a')$ be their first common ancestor toward its root and put

$$F_T(a, a') = [a, q_T(a, a')]_T \cup [a', q_T(a, a')]_T.$$

Define the global SDP cost

$$J_T(a, a') = \max\{d(u, v) : [u, v]_T \subset F_T(a, a') \text{ is map-directed}\}. \tag{2.7}$$

The distance in (2.7) is computed in the whole map M, not in the tree.

Lemma 2.3 (Contour intervals)**.** *Fix $T \in \mathcal{T}$ and a finite integer interval $I = [a, b] \cap \mathbb{Z}$ in its associated clock. The first common ancestor of all black points with labels in I is the common ancestor of the two endpoint points. Denote it by $q_T(I)$ and put*

$$H_T(I) = \bigcup_{k \in I} [\lambda^{c(T)}(k), q_T(I)]_T. \tag{2.8}$$

The sum of the intrinsic lengths of the two endpoint branches, with half-edges of length $1/2$, is at most

$$C_0(1 + \operatorname{osc}_I Z^{c(T)}). \tag{2.9}$$

If $I' \subset I$, then $H_T(I') \subset H_T(I)$. If I' is one of the two children in a bisection of I, then $[q_T(I'), q_T(I)]_T$ lies on the endpoint branch of I corresponding to the endpoint shared by I' and I.

Proof. Cutting all non-selected half-edges turns the relevant extremal tree into a plane tree with its contour order; this is the cut-tree construction in [6, Sections 1.2 and 2.1]. Descendants of a fixed tree vertex occur consecutively in that order. Hence the first common ancestor of the two endpoints is an ancestor of every visit between them, proving the first assertion and monotonicity.

Equivalently, in the stack description the ancestor level is the minimum of the relevant height coordinate between the two endpoint visits. Distances to this minimum are bounded by that coordinate's oscillation. Restoring endpoint half-edges changes lengths by at most two, proving (2.9). The same description gives coalescence in the infinite tree.

If I' and I share endpoint k, both ancestors lie on the branch from $\lambda^{c(T)}(k)$, with $q_T(I)$ farther toward the root. This is the last assertion. □

2.3 Planar intersections and source transfer

We first work in a finite plane bipolar map. Subdivide multiple edges first, so that an upward embedding may be used. Directed paths can then be represented as continuous horizontal-coordinate functions of height, after a harmless perturbation making vertex heights distinct.

For a point a, write r_a^+, ℓ_a^+ for its rightmost and leftmost outgoing paths to the sink. They are pointwise extremal among directed paths from a to the sink. Let r_b^- be the rightmost source-to-b path, and define the other incoming extreme analogously.

Lemma 2.4 (Two planar intersections)**.** *In a finite plane bipolar map, for black points a, b, a':*

(i) If $a \preceq b$, then $[a, q_{T_R^+}(a, b)]_{T_R^+}$ intersects $[b, q_{T_R^-}(a, b)]_{T_R^-}$.

(ii) Suppose $a \preceq b$, $a' \preceq b$, and $a' \not\preceq a$. Every directed path from a to b intersects

$$[a', q_{T_R^+}(a, a')]_{T_R^+} \cup [a', q_{T_L^+}(a, a')]_{T_L^+}. \tag{2.10}$$

Proof. For (i), extend an a-to-b path to the source and sink. At the height of a, the rightmost source-to-b path lies weakly to the right of the rightmost a-to-sink path; at the height of b the latter lies weakly to the right of the former. Continuity gives an intersection at an intermediate height. The outgoing common ancestor lies above b and the incoming common ancestor below a, so the intersection belongs to both asserted prefixes.

For (ii), the vertices reachable from a' are exactly the vertices at height at least that of a' in the closed region between $\ell^+_{a'}$ and $r^+_{a'}$. Let P be an a-to-b path. Since $a' \not\preceq a$ but $a' \preceq b$, P has a first entrance into this region. If the entrance is at the height of a', it is a'. Otherwise suppose, for example, that P enters through $r^+_{a'}$. If the entrance were strictly beyond the coalescence point of r^+_a and $r^+_{a'}$, then immediately below it the two rightmost paths would coincide while P lay strictly to their right, contradicting extremality of r^+_a. Thus the entrance lies on the first branch in (2.10). The left side is identical. Pinches and cut vertices cause no problem; after subdivision every intersection is a graph vertex. □

Lemma 2.5 (Infinite-volume transfer and the two orders)**.** *Almost surely, Lemma 2.4 holds in M, and (1.3) holds. Moreover,*

$$d(a,b) \le J_{T_R^+}(a,b) + J_{T_R^-}(a,b), \qquad a \preceq b, \tag{2.11}$$

and, with $J(a,a') = \sum_{T\in\mathcal{T}} J_T(a,a')$,

$$|d(a,b) - d(a',b)| \le J(a,a'), \qquad a \preceq b,\ a' \preceq b, \tag{2.12}$$

$$|d(a,b) - d(a,b')| \le J(b,b'), \qquad a \preceq b,\ a \preceq b'. \tag{2.13}$$

Proof. The UIBOT is supported on the local closure of finite plane bipolar triangulations; see [2, Definition 3.9 and Proposition 3.10]. Work on a realization for which all four relevant tree branches coalesce for every pair of marked points.

We first record the stabilization fact needed below. A finite segment of either primal Peano order is determined by finitely many local successor decisions. Each successor decision is determined by the plane incidence order in the full star of the current vertex together with the current edge. Therefore, for any finite set of marked edges and any finite N- or W-exploration segment joining them to the root, there is a finite vertex set containing the full stars encountered by that segment such that every finite bipolar approximant agreeing on this finite rooted neighborhood has exactly the same marked exploration segment, with the same relative labels. This is a deterministic statement about the Peano successor rule; no global completion information is used.

Now fix a finite marked set. Enlarge a finite neighborhood so that it contains the four coalescence prefixes of each relevant pair, the full incident stars of all vertices on those prefixes, and any particular finite directed paths under consideration. Finite approximating maps then preserve the prefixes and their first coalescences. A finite approximant cannot create a comparability between two marked points which is absent in M: by Lemma 2.4(i), such a comparability would force an intersection of the preserved outgoing/incoming extremal prefixes, and this same intersection already gives a directed path in M. It follows that the case distinction in Lemma 2.4(ii) is also preserved. Passing Lemma 2.4 from the finite approximants proves its infinite-volume form.

In a finite plane bipolar map, the Baxter/BBMF construction identifies reachability of black points with coordinatewise comparison of the two exploration orders; see [1, Proposition 1 and Theorem 2] and [3, Definition 1.3 and Section 2.3]. The stabilization fact in the first paragraph preserves the two relative exploration labels of any fixed marked pair in the finite approximants. If $p(a) \le p(b)$ there, the finite product-order theorem gives $a \preceq b$ in the approximant; Lemma 2.4(i), applied in the approximant, forces an intersection of the preserved extremal prefixes, hence a directed path from a to b in M. Conversely, if $a \preceq b$ in M, then both primal Peano labels

increase along the directed path by Lemma 2.1 and its reflected version. This proves (1.3). A countable intersection handles all marked pairs.

For (2.11), choose the intersection from Lemma 2.4(i), join a to it and it to b by global shortest paths, and use the triangle inequality. For one direction of (2.12), if $a' \preceq a$ then

$$d(a', b) - d(a, b) \le d(a', a) \le J(a, a').$$

Otherwise apply Lemma 2.4(ii) to a shortest a-to-b path, join a' to its first intersection with (2.10), and retain the old terminal segment. The extra cost is at most $J(a, a')$. Interchanging a, a' gives the absolute value. Reversing all orientations gives (2.13); this merely permutes the four trees. □

3 An exponential boundary-segment estimate

Let $M_{0,\infty}$ be the future half-map in the N-encoding, and let $x_0, x_1, \dots$ be its right-boundary vertices in directed order, with root edge $x_0 \to x_1$. Define

$$B_r = \max_{0 \le a \le b \le r} \mathrm{SDP}_{M_{0,\infty}}(x_a, x_b), \qquad r \ge 1. \tag{3.1}$$

In this section only, all distances in (3.1) are half-map distances.

Proposition 3.1 (Boundary tail)**.** *There are $c, C > 0$ such that*

$$\mathbb{P}[B_r > A r^{3/4}] \le C e^{-cA}, \qquad r \ge 1,\ A \ge 1. \tag{3.2}$$

3.1 A uniform upper quantile from the marked cap

We use two results of [2]. First, [2, Lemma 7.8] says that for every $\eta > 0$ there is $K_\eta < \infty$ such that, uniformly in $N \ge 1$, with probability at least $1 - \eta$, every lower-boundary vertex of $M_{0,N}$ can be joined inside $M_{0,N}$ to some upper-boundary vertex by a directed path of length at most $K_\eta N^\sigma$.

Second, we use the marked cap of [2, Lemma 7.9]. Let $\widehat{M}_{0,\infty}$ be the submap of $M_{0,\infty}$ induced by vertices reachable from x_0; it has the UIQBOT law. For each $r \ge 1$ there is a finite complete cap $\widehat{M}(r) \subset \widehat{M}_{0,\infty}$ with source x_0, sink x_r, right boundary $x_0, \dots, x_r$, and a marked left-boundary vertex $z(r)$. Write I_r for the left-boundary segment from x_0 to $z(r)$ and J_r for the left-boundary segment from $z(r)$ to x_r. Its marked Boltzmann law is invariant under reversing all orientations and then applying an orientation-reversing homeomorphism. If

$$\tau_0 = 0, \qquad \tau_r = \inf\{t \ge 0 : R(t) = -r\}, \tag{3.3}$$

then the coupling in [2, (7.32)–(7.37)] gives

$$\widehat{M}(r) \subset M_{0,\tau_r - 1}. \tag{3.4}$$

Only this inclusion is used below; the cap is not identified with an unconditioned stopped KMSW cell.

Lemma 3.2 (Cap separation)**.** *Fix $r \ge 1$ and N such that $\widehat{M}(r) \subset M_{0,N}$. Let $0 \le i \le r$. Every directed path in $M_{0,N}$ from x_i to the upper boundary of $M_{0,N}$ intersects J_r. The segment before its first intersection with J_r is contained in $\widehat{M}(r)$.*

Proof. Let P be such a path. Concatenate to P the directed right-boundary segment from x_0 to x_i. This gives a directed path $\widetilde{P}$ from x_0 to the upper boundary of $M_{0,N}$. The planar-separation argument in [2, Lemma 7.13, Step 1], whose only geometric input is $\widehat{M}(r) \subset M_{0,N}$, shows that every such x_0-to-upper-boundary path meets J_r. The added right-boundary segment $x_0, \dots, x_i$ is disjoint from J_r unless $i = r$, in which case it meets J_r only at the common endpoint x_r. Hence P itself meets J_r.

For the second assertion, recall the cap geometry from [2, Lemma 7.9]: in the induced reachable map $\widehat{M}_{0,\infty}$, the arc J_r, together with the right-boundary segment ending at x_r, separates the finite cap from its future side. Since $x_0 \preceq x_i$, the path P lies in $\widehat{M}_{0,\infty}$. It therefore cannot leave the cap before meeting J_r. If the boundary has repeated vertices or pinches, take a sufficiently small regular neighborhood of the embedded finite submap; the same assertion becomes the ordinary crosscut separation statement in a topological disk, and projecting back counts a shared endpoint as an intersection. □

Lemma 3.3 (Boundary upper quantile)**.** *There is $K < \infty$ such that*

$$\sup_{r \geq 1} \mathbb{P}[B_r > K r^{3/4}] \leq \frac{1}{4}. \tag{3.5}$$

Proof. Fix $\eta > 0$. A standard hitting-time estimate for the centered lazy walk R gives $K_0 < \infty$ such that

$$\sup_{r \geq 1} \mathbb{P}[\tau_r > K_0 r^2] \leq \eta.$$

Set $N = \lceil K_0 r^2 \rceil$. On $\{\tau_r \leq N\}$, (3.4) gives $\widehat{M}(r) \subset M_{0,N}$, and $x_0, \dots, x_r$ are lower-boundary vertices of $M_{0,N}$.

Apply [2, Lemma 7.8]. Except on an event of probability at most η, simultaneously for every $0 \leq i \leq r$ there is a directed path in $M_{0,N}$ from x_i to the upper boundary with length at most $K_\eta N^\sigma$. By Lemma 3.2, truncating at the first hit of J_r yields, simultaneously for all i,

$$x_i \longrightarrow J_r \quad \text{inside } \widehat{M}(r), \qquad \text{length} \leq K_1 r^{3/4}, \tag{3.6}$$

for a deterministic K_1 depending only on η and K_0.

The event in (3.6) is measurable from the marked cap alone. Under the reversal/reflection involution of [2, Lemma 7.13, Step 2], the source and sink are interchanged, I_r and J_r are interchanged, and the right-boundary vertex x_i is sent to x_{r-i}. Thus the image of (3.6) is exactly the event that, simultaneously for all $0 \leq j \leq r$,

$$I_r \longrightarrow x_j \quad \text{inside } \widehat{M}(r), \qquad \text{length} \leq K_1 r^{3/4}. \tag{3.7}$$

Since the marked-cap law is invariant under this involution, (3.7) has the same probability as (3.6), hence at least $1 - 2\eta$. A union bound therefore gives probability at least $1 - 4\eta$ for their simultaneous occurrence. No independence is asserted or needed.

Fix $i < j$. Choose paths as in (3.6) and (3.7). Their endpoint arcs alternate on the boundary of the cap, so the two paths intersect. With pinches, the same conclusion follows after passing to a small regular neighborhood, as in Lemma 3.2. Concatenating the initial part of the first path with the terminal part of the second gives

$$\mathrm{SDP}_{\widehat{M}(r)}(x_i, x_j) \leq 2K_1 r^{3/4}.$$

The distance in $M_{0,\infty}$ is no larger. Choose $\eta \leq 1/16$ and enlarge the constant to absorb integer rounding. This proves (3.5). □

3.2 Independent excursions and disjoint certificates

For $k \geq 1$, let E_k be the finite vector-valued word of increments of Z between τ_{k-1} and τ_k. By the strong Markov property and spatial translation invariance, the words $E_1, E_2, \ldots$ are i.i.d. with values in a countable alphabet.

Lemma 3.4 (Excursion locality)**.** *For $0 \leq a < b$, the distance $\mathrm{SDP}_{M_{0,\infty}}(x_a, x_b)$ is determined by $E_{a+1}, \ldots, E_b$ and is unchanged when all other excursion words are resampled. In particular, B_r is a function of $E_1, \ldots, E_r$.*

Proof. The boundary-time identity [2, (5.2)–(5.3)] says that τ_a is the first active time of an edge leaving x_a. The head-to-tail phase separation in Lemma 2.1 implies that every incoming edge at x_b has active time strictly before τ_b. Hence every directed path from x_a to x_b uses only active labels in $[\tau_a, \tau_b - 1]$.

The endpoints are read from the anchor edges at τ_a and τ_b. Lemma 2.1, applied to $[\tau_a, \tau_b]$, shows that all signed incidences among the relevant edges depend only on the increments in $(\tau_a, \tau_b]$, namely $E_{a+1}, \ldots, E_b$. Resampling an earlier excursion changes absolute active labels but not this relative word or the boundary indices; a later excursion cannot supply an incoming edge at x_b. The distance is therefore unchanged. □

Lemma 3.5 (Boundary recursion)**.** *For integers $r, k, h \geq 1$,*

$$\mathbb{P}[B_r \geq kh] \leq \mathbb{P}[B_r \geq h]^k. \tag{3.8}$$

Proof. On $\{B_r \geq kh\}$ choose $i < j$ with $q = \mathrm{SDP}_{M_{0,\infty}}(x_i, x_j) \geq kh$, and set

$$F(a) = \mathrm{SDP}_{M_{0,\infty}}(x_a, x_j), \qquad i \leq a \leq j.$$

The boundary edge $x_a \to x_{a+1}$ gives $F(a+1) \geq F(a) - 1$. Since $F(i) = q$ and $F(j) = 0$, the first downward crossings of $q - h, q - 2h, \ldots, q - kh$ hit these levels exactly. Let their indices be $a_1 < \cdots < a_k$ and put $a_0 = i$. The directed triangle inequality gives

$$\mathrm{SDP}_{M_{0,\infty}}(x_{a_{\ell-1}}, x_{a_\ell}) \geq F(a_{\ell-1}) - F(a_\ell) = h.$$

By Lemma 3.4, fixing the excursion coordinates with indices in $(a_{\ell-1}, a_\ell] \cap \mathbb{Z}$ forces $\{B_r \geq h\}$ regardless of the other excursion coordinates. These k coordinate sets are disjoint. Hence $\{B_r \geq kh\}$ is contained in the k-fold disjoint occurrence of $\{B_r \geq h\}$, and Reimer's inequality gives (3.8) when the excursion alphabet is finite.

For the countable alphabet, choose increasing finite subsets whose probabilities tend to one, condition each of the first r coordinates to lie in the corresponding finite set, and apply the finite-product inequality. The certificate inclusion is unchanged under this restriction, and the resulting product measures converge in total variation to the original law on the first r coordinates. Passing to the limit gives (3.8). □

Proof of Proposition 3.1. Let $h_r = \lfloor K r^{3/4} \rfloor + 1$, with K from Lemma 3.3. Then $\mathbb{P}[B_r \geq h_r] \leq 1/4$ and $h_r \leq (K+1) r^{3/4}$. For $A \geq 2(K+1)$ take $k = \lfloor A/(K+1) \rfloor$. Since $k h_r \leq A r^{3/4}$, Lemma 3.5 gives

$$\mathbb{P}[B_r > A r^{3/4}] \leq \mathbb{P}[B_r \geq k h_r] \leq 4^{-k} \leq C e^{-cA}.$$

Increasing C handles bounded A. □

4 Global SDP costs on tree intervals

For $T \in \mathcal{T}$ and a nonempty finite interval I in its associated clock, define

$$S_T(I) = \max\{d(u,v) : [u,v]_T \subset F_T(\lambda(a), \lambda(b)) \text{ is map-directed}\}, \tag{4.1}$$
$$C_T(I) = \max\{d(u,v) : [u,v]_T \subset H_T(I) \text{ is map-directed}\}, \tag{4.2}$$

where a, b are the endpoints of I and $\lambda = \lambda^{c(T)}$. Both costs use the global metric d_M. By Lemma 2.3, $C_T(I)$ is monotone under inclusion of intervals. In particular,

$$J_T(a, a') \le C_T(I_{c(T)}(a, a')), \tag{4.3}$$

where $I_N(a, a')$ and $I_W(a, a')$ are the integer intervals between the corresponding clock labels.

Lemma 4.1 (Extremal rays versus half-map boundaries)**.** *There is a universal C_1 with the following property. Fix $T \in \mathcal{T}$ and a deterministic time s in its associated clock. If a branch segment of T starting at $\lambda^{c(T)}(s)$ has intrinsic length at most r, then its global SDP cost is stochastically dominated by $B_{\lceil C_1 r \rceil + C_1} + C_1$.*

Proof. For T_R^+ in the N clock, the branch from a deterministic active edge toward the future consists, after the endpoint half-edge adjustment, of the successive rightmost outgoing edges. In the future half-map rooted at that active edge these are successive edges of the directed right boundary. Thus a directed subsegment of a branch of intrinsic length at most r is contained in a boundary segment with at most $C_1 r + C_1$ boundary steps. Its distance in the whole map is no larger than its distance in the half-map, so (3.1) gives the asserted domination.

The other three cases are obtained by deterministic symmetries of the sewing. Left-right reflection sends T_R^+ to T_L^+ and replaces the N exploration by the primal W exploration. By [5, Remark 1.8], the latter has the same i.i.d. three-step law. Reversing all orientations exchanges future and past trees, and the induced time reversal, together with swapping the two walk coordinates, again permutes the three allowed increments. Therefore the same boundary law and the same deterministic endpoint constant apply to T_L^+, T_R^-, T_L^-. No dual encoding is assigned the i.i.d. law in this argument. □

Proposition 4.2 (Tree-interval tail)**.** *For every $T \in \mathcal{T}$, every deterministic nonempty integer interval I, and $A \ge 1$,*

$$\mathbb{P}[C_T(I) > A|I|^\sigma] \le Ce^{-c\sqrt{A}}. \tag{4.4}$$

Proof. Let $m = |I|$. We first estimate the two endpoint branches comprising $F_T(\lambda(a), \lambda(b))$. By Lemma 2.3, their total intrinsic length is bounded by $C(1 + \operatorname{osc}_I Z^{c(T)})$. Since the relevant encoding walk has bounded centered increments, a maximal inequality gives

$$\mathbb{P}[\text{an endpoint branch has intrinsic length} > r] \le Ce^{-cr^2/m}$$

for $r \ge C\sqrt{m}$; after changing constants the bound is harmless for all $r \ge 1$. On the complementary event, Lemma 4.1 and Proposition 3.1 bound the SDP cost of each endpoint branch. A directed tree segment in the union of the two branches cannot traverse through the common ancestor from one branch to the other, since both branches are oriented in the same direction relative to that ancestor. Hence, for $r \ge 1$,

$$\mathbb{P}[S_T(I) > Am^\sigma] \le Ce^{-cr^2/m} + C\exp\left\{-c\frac{Am^\sigma}{(r+2)^{3/4}}\right\}. \tag{4.5}$$

The two events in this union bound need not be independent. Take r to be a sufficiently large constant times $\lceil A^{2/3}\sqrt{m}\rceil$. Since $\sigma = 3/8$, the second exponent is of order $A^{1/2}$ and the first is stronger. Thus

$$\mathbb{P}[S_T(I) > Am^\sigma] \le Ce^{-c\sqrt{A}}. \tag{4.6}$$

Assume next that m is a power of two and bisect I down to singletons. Let $\mathcal{D}_\ell(I)$ be the level-ℓ intervals, with $\ell = 0$ at the root. Fix $k \in I$ and follow the nested chain from I to $\{k\}$. The last assertion of Lemma 2.3 puts every successive ancestor connector on an endpoint branch of its parent interval. These connectors partition the branch from $\lambda^{c(T)}(k)$ to $q_T(I)$. Therefore every map-directed tree segment in $H_T(I)$ can be split into at most one map-directed piece from a connector at each level, and each such piece has cost at most $S_T(J)$ for the corresponding interval J. Consequently

$$C_T(I) \le \sum_{\ell \ge 0} \max_{J \in \mathcal{D}_\ell(I)} S_T(J). \tag{4.7}$$

At level ℓ, use threshold $Am^\sigma 2^{-\sigma\ell/2}$. Each $J \in \mathcal{D}_\ell(I)$ has cardinality $m2^{-\ell}$, so the threshold divided by $|J|^\sigma$ equals $A2^{\sigma\ell/2}$. By (4.6) and a union bound,

$$\mathbb{P}\left[\max_{J \in \mathcal{D}_\ell(I)} S_T(J) > Am^\sigma 2^{-\sigma\ell/2}\right] \le C2^\ell \exp\{-c\sqrt{A}\, 2^{\sigma\ell/4}\}.$$

The sum over ℓ is at most $C'e^{-c'\sqrt{A}}$, while the thresholds sum to a constant multiple of Am^σ. This proves (4.4) for power-of-two cardinality, after changing constants. Every finite interval is contained in a deterministic interval of power-of-two cardinality at most $2m$; use monotonicity to conclude. □

Define

$$\begin{aligned} K(a,a') = {} & C_{T_R^+}(I_N(a,a')) + C_{T_L^-}(I_N(a,a')) \\ & + C_{T_L^+}(I_W(a,a')) + C_{T_R^-}(I_W(a,a')). \end{aligned} \tag{4.8}$$

By (4.3), the comparisons (2.12)–(2.13) hold with K in place of J. Also, by (2.11),

$$d(a,b) \le C_{T_R^+}(I_N(a,b)) + C_{T_R^-}(I_W(a,b)), \qquad a \preceq b. \tag{4.9}$$

5 Transverse compactness for a whole cell

This section proves a uniform statement for all edges of one deterministic N-block. Tightness of the transverse label of a single deterministic edge is not sufficient, because a cell-diameter maximizer is selected adaptively from the whole block.

5.1 The GHS north and south regions

For deterministic $t \ge 1$, let $M_{0,t}^N$ and $M_{0,t}^S$ be the north and south edge sets of [5, Section 5.1], defined using the dual north-going and south-going flow lines from the ordinary Peano edges $\lambda^N(0)$ and $\lambda^N(t)$, including the edge-side convention used there. Put

$$Q_t^+ = \#M_{0,t}^N, \qquad Q_t^- = \#M_{0,t}^S, \qquad Q_0^+ = Q_0^- = 0.$$

Lemma 5.1 (Transverse-region decomposition)**.** *There is a deterministic $C_0 < \infty$ such that, for every $t \geq 0$,*

$$|j(\lambda^N(t))| \leq t + 1 + Q_t^+ + Q_t^- + C_0. \tag{5.1}$$

Proof. For $t \geq 1$, let $\mathcal{R}_t$ be the finite edge region disconnected from infinity by the four dual flow lines $\lambda_0^N, \lambda_0^S, \lambda_t^N, \lambda_t^S$ in the notation of [5, Section 5.1]. As observed immediately before [5, Proposition 5.1], the amount of dual west-going Peano time between the edge crossing $\lambda^N(0)$ and the edge crossing $\lambda^N(t)$ is exactly $\#\mathcal{R}_t$. By [5, Lemma 2.5], the dual west-going and primal west-going Peano curves visit edges in the same order. With both orders rooted at the same edge, this gives

$$|j(\lambda^N(t))| \leq \#\mathcal{R}_t + C_0,$$

where C_0 only allows for the harmless choice of whether a boundary-crossing edge is counted on one side or the other.

Classify an edge of $\mathcal{R}_t$ by its ordinary N-Peano time. There are at most $t+1$ edges with time in $[0, t]$. By the definitions in [5, Section 5.1], an edge of the four-flow-line enclosure whose ordinary Peano time is $> t$ lies on the future side of the explored strip and between the two north-going flow lines; with the edge-side convention of that section it is therefore an element of $M_{0,t}^N$. The time-reversed statement says that an enclosed edge with ordinary Peano time < 0 belongs to $M_{0,t}^S$. We only need this inclusion, not a disjoint partition. Hence

$$\mathcal{R}_t \subset \lambda^N([0, t] \cap \mathbb{Z}) \cup M_{0,t}^N \cup M_{0,t}^S$$

up to the bounded convention for boundary-crossing edges, and consequently

$$\#\mathcal{R}_t \leq t + 1 + Q_t^+ + Q_t^- + C_0.$$

After enlarging C_0 once, this proves (5.1). The case $t = 0$ is immediate. □

Let H_d be the first hitting time of -1 by a lazy random walk with increments uniform on $\{-1, 0, 1\}$, started from $d \in \mathbb{N}_0$. We will use the standard estimate

$$\mathbb{P}[H_d > u] \leq C \min\{1, (d+1)u^{-1/2}\}, \qquad d \geq 0,\ u \geq 1. \tag{5.2}$$

One way to prove (5.2) is to delete zero increments, apply the reflection principle to the resulting simple random walk, and use an exponential lower-tail bound for the number of nonzero increments.

Set

$$\mathcal{F}_t = \sigma(Z(s) : s \leq t), \qquad \mathcal{G}_0 = \sigma(Z(s) : s \geq 0).$$

Lemma 5.2 (Two-sided conditional pocket bounds)**.** *For deterministic integers $t \geq 0$ and $u \geq 1$,*

$$\mathbb{P}[Q_t^+ > u \mid \mathcal{F}_t] \leq C \min\{1, (1 + \operatorname{osc}_{[0,t]} Z)u^{-1/2}\}, \tag{5.3}$$

$$\mathbb{P}[Q_t^- > u \mid \mathcal{G}_0] \leq C \min\{1, (1 + \operatorname{osc}_{[0,t]} Z)u^{-1/2}\}. \tag{5.4}$$

The inequalities are understood as almost-sure inequalities between versions of conditional probabilities, and the versions may be chosen simultaneously for the countable set of integer pairs (t, u).

Proof. For the north region, [5, Lemma 5.2] states that, conditionally on $\mathcal{F}_t$, Q_t^+ is the first hitting time of -1 by a lazy walk started from the exposed boundary separation d_t^+. In the proof of [5, Proposition 5.1], this separation is bounded deterministically by $|Z(t)|$ (with their normalization). Since $Z(0) = 0$,

$$d_t^+ \le |Z(t)| \le C \operatorname{osc}_{[0,t]} Z.$$

Allowing a fixed endpoint convention gives the displayed $1 + \operatorname{osc}$ bound. Applying (5.2) proves (5.3).

For completeness, we derive the conditioning in the south estimate rather than merely invoking an unconditional symmetry. Let $\iota(x, y) = (y, x)$ and, for fixed t, define the reversed encoding

$$\check{Z}_t(s) = \iota\big(Z(t - s) - Z(t)\big), \qquad s \in \mathbb{Z}.$$

Then

$$\check{Z}_t(s) - \check{Z}_t(s - 1) = -\iota\big(Z(t - s + 1) - Z(t - s)\big).$$

The map $v \mapsto -\iota(v)$ fixes $X = (1, -1)$ and interchanges $L_0 = (-1, 0)$ with $R_0 = (0, 1)$. Hence $\check{Z}_t$ has the same i.i.d. increment law as Z. Reversing orientations and reflecting the bipolar map transforms the original south region $M_{0,t}^S$ into the north region between times 0 and t for $\check{Z}_t$. Moreover,

$$\sigma(\check{Z}_t(s) : s \le t) = \sigma(Z(u) - Z(0) : u \ge 0) = \mathcal{G}_0,$$

where the deterministic value $Z(0) = 0$ is irrelevant. The exposed separation for the reversed north region is bounded by $C(1 + \operatorname{osc}_{[0,t]} Z)$. Applying [5, Lemma 5.2] to $\check{Z}_t$, followed by (5.2), proves (5.4). □

Lemma 5.3 (Stopped transverse labels)**.** *Let T_n be nonnegative integer-valued stopping times for the filtration generated by the increments of Z at nonnegative times. Suppose that both families*

$$\{T_n/n : n \ge 1\}, \qquad \{n^{-1/2} \operatorname{osc}_{[0,T_n]} Z : n \ge 1\} \tag{5.5}$$

are tight. Then $\{n^{-1} j(\lambda^N(T_n)) : n \ge 1\}$ is tight.

Proof. Fix $\eta > 0$. Choose $K, B < \infty$ such that

$$E_n = \{T_n \le Kn,\ \operatorname{osc}_{[0,T_n]} Z \le B\sqrt{n}\}$$

has probability at least $1 - \eta$, uniformly in n after increasing K, B if necessary.

For the future region, $\{T_n = t\} \cap E_n \in \mathcal{F}_t$. Summing (5.3) over the possible values of T_n gives, for $A \ge 1$,

$$\mathbb{P}[Q_{T_n}^+ > An, E_n] \le C(B + 1)A^{-1/2}.$$

For the past region, both T_n and E_n are $\mathcal{G}_0$-measurable. Using the simultaneous versions in Lemma 5.2 and summing (5.4) over the now-known value of T_n gives

$$\mathbb{P}[Q_{T_n}^- > An, E_n] \le C(B + 1)A^{-1/2}.$$

Finally, on E_n we have $T_n \le Kn$, so (5.1) implies tightness after first choosing A large and then η small. No reversed stopping-time assertion is used. □

5.2 Enclosing a deterministic block

Lemma 5.4 (Record-edge enclosure)**.** *For $r \geq 3$, set*

$$T^+(r) = \inf\{t \geq 0 : R(t) = -r\}, \qquad T^-(r) = -\inf\{s \geq 0 : L(-s) = -r\},$$

and $e^+(r) = \lambda^N(T^+(r))$, $e^-(r) = \lambda^N(T^-(r))$. There are universal constants $c, C > 0$ such that, for every $n \geq 1$ and $r \geq C$,

$$\mathbb{P}\big[e^-(r) \not\preceq \lambda^N(k) \text{ or } \lambda^N(k) \not\preceq e^+(r) \text{ for some } 0 \leq k \leq n\big] \leq C \exp\{-cr^2/n\}, \tag{5.6}$$

with the right side interpreted as a harmless constant when $r^2 < n$.

Proof. We prove the future assertion; the past assertion is its reversed/reflected counterpart. The boundary-time identity [2, (5.2)–(5.3)], equivalently the tail-frontier stack description, identifies successive times at which R attains a new negative record with successive edges on the rightmost outgoing branch from the root, up to the fixed root half-edge convention. Thus $e^+(r)$ lies at tree depth $r + O(1)$ on the root branch toward the future.

Let $q = q_{T_R^+}([0, n])$ be the first common ancestor of the block edges $\lambda^N(0), \dots, \lambda^N(n)$. By Lemma 2.3, the depth of q below $\lambda^N(0)$ is at most $C_0(1 + \operatorname{osc}_{[0,n]} Z)$. If this depth is less than $r - C_1$, then $e^+(r)$ lies strictly beyond q toward the root of the future tree. Every block edge therefore reaches $e^+(r)$ along its directed T_R^+ ancestor chain, so $\lambda^N(k) \preceq e^+(r)$ for all $k \in [0, n] \cap \mathbb{Z}$.

For the past side, apply the same argument after reversing orientations and reflecting the map. The relevant contour coordinate becomes the backward L-coordinate, and one obtains $e^-(r) \preceq \lambda^N(k)$ for every k whenever the analogous incoming common-ancestor depth is less than $r - C_1$. The complement of the two depth events is contained in

$$\{\operatorname{osc}_{[0,n]} Z \geq c_1 r - C_2\}.$$

A maximal Hoeffding bound for the two bounded centered coordinate walks gives (5.6), after changing constants. □

Proposition 5.5 (Block transverse compactness)**.** *The family*

$$\left\{ \frac{1}{n} \max_{0 \leq k \leq n} |j(\lambda^N(k))| : n \geq 1 \right\} \tag{5.7}$$

is tight.

Proof. Fix $B > 0$ and put $r_n = \lceil B\sqrt{n} \rceil + 3$. Define

$$T_n^+ = \inf\{t \geq 0 : R(t) = -r_n\}, \qquad T_n^- = -\inf\{s \geq 0 : L(-s) = -r_n\},$$

and set $e^+ = \lambda^N(T_n^+)$ and $e^- = \lambda^N(T_n^-)$. Lemma 5.4, with $r = r_n$, gives

$$e^- \preceq \lambda^N(k) \preceq e^+, \qquad 0 \leq k \leq n, \tag{5.8}$$

except on an event whose probability is at most Ce^{-cB^2} after changing constants. In particular,

$$\lim_{B \to \infty} \sup_{n \geq 1} \mathbb{P}[(5.8) \text{ fails}] = 0. \tag{5.9}$$

For fixed B, the hitting-time estimate (5.2) implies that T_n^+/n is tight. Its stopped oscillation divided by $\sqrt{n}$ is also tight: restrict first to $T_n^+ \le Kn$ and then use a deterministic-time maximal inequality on $[0, Kn]$. Lemma 5.3 therefore shows that $j(e^+)/n$ is tight.

Apply the same argument to the map obtained by reversing all orientations and reflecting, encoded by $\check{Z}_0(s) = \iota(Z(-s) - Z(0))$. Then $-T_n^-$ is a stopping time for its nonnegative-time filtration. Both primal Peano orders are defined by local successor rules, so this involution reverses each order: after re-rooting its zero edge, the transformed transverse label of the image of e^- is $-j(e^-)$, up to the bounded root-edge indexing convention. Lemma 5.3 in the transformed map therefore implies tightness of $j(e^-)/n$.

On (5.8), the W labels increase along directed paths by (1.3), so

$$j(e^-) \le j(\lambda^N(k)) \le j(e^+), \qquad 0 \le k \le n.$$

Choose B so that the failure probability in (5.9) is small and then use the tightness of the two endpoint labels. This proves (5.7). □

6 Ordinary-cell tails and the median scale

For a finite interval I in either primal clock $\star \in \{N, W\}$, put

$$A^\star(I) = \max\{d(e, f) : e, f \in \lambda^\star(I),\ e \preceq f\}. \tag{6.1}$$

This is monotone under inclusion of I. By endpoint locality, anchoring the first and last active edges confines every admissible path to their clock interval, so $A^\star(I)$ is determined by the active-edge graph of that interval.

Lemma 6.1 (Anchored and vertex diameters)**.** *For the N-cell,*

$$|\Delta_n - A^N([0, n])| \le 1. \tag{6.2}$$

The same assertion holds for a cell in the W-encoding.

Proof. A vertex geodesic of length Δ_n supplies its first and last edge midpoints. The distance between those midpoints is $\Delta_n - 1$: a shorter anchored path would shorten the vertex geodesic, and Lemma 2.1 confines any global anchored shortcut to the cell. Hence $A^N([0, n]) \ge \Delta_n - 1$.

Conversely, for distinct comparable cell edges e, f, the midpoint distance is one plus the shortest distance between the head of e and the tail of f. Every path in this middle expression is confined between the two anchor labels, and therefore lies in the cell. Thus $d(e, f) \le \Delta_n + 1$. The case $e = f$ is immediate. □

Proposition 6.2 (A fixed upper quantile)**.** *There is $K < \infty$ such that*

$$\sup_{n \ge 0} \mathbb{P}[\Delta_n > K(n+1)^\sigma] \le \frac{1}{4}. \tag{6.3}$$

Proof. By Proposition 5.5, for any prescribed small $\eta > 0$ we can choose K_0 such that, uniformly in n,

$$\max_{0 \le k \le n} |j(\lambda^N(k))| \le K_0 n$$

except on an event of probability at most η (enlarging K_0 handles finitely many small n). On this event, (4.9) and monotonicity give

$$A^N([0,n]) \le C_{T_R^+}([0,n]) + C_{T_R^-}([-K_0 n, K_0 n] \cap \mathbb{Z}).$$

Proposition 4.2 makes the right side at most a deterministic multiple of $(n+1)^\sigma$ with arbitrarily high fixed probability. Choose the two error probabilities to total at most $1/4$, then use Lemma 6.1 and enlarge the constant. □

Proof of Theorem 1.2. Let $h_n = \lfloor K(n+1)^\sigma \rfloor + 1$, with K from Proposition 6.2. Then $\mathbb{P}[\Delta_n \ge h_n] \le 1/4$ and

$$h_n + 2 \le (K+3)(n+1)^\sigma.$$

For $A \ge 2(K+3)$ take $k = \lfloor A/(K+3) \rfloor$. Since $k(h_n+2) \le A(n+1)^\sigma$, Lemma 2.2 gives

$$\mathbb{P}[\Delta_n > A(n+1)^\sigma] \le \mathbb{P}[\Delta_n \ge k(h_n+2)] \le 4^{-k} \le Ce^{-cA}.$$

This proves (1.7), after increasing C for bounded A.

For the lower median bound we use exactly the third item of [2, Theorem 1.13]. Taking its error probability to be $1/4$, there is $c_0 > 0$ such that for every $n \ge 1$, with probability at least $3/4$ there are two vertices of $M_{0,n}$ at SDP distance at least $c_0 n^\sigma$. Therefore

$$\mathbb{P}[\Delta_n \ge c_0 n^\sigma] \ge \frac{3}{4}. \tag{6.4}$$

Together, (6.3) and (6.4) imply $cn^\sigma \le m_n \le Cn^\sigma$ for every choice of medians. This proves (1.8), with no monotonicity or regular-variation assumption on m_n.

Finally, the N and primal W encodings have identical marginal laws by [5, Remark 1.8]. Lemma 6.1 and (1.7), applied after translation of a deterministic interval, give

$$\mathbb{P}[A^\star(I) > A|I|^\sigma] \le Ce^{-cA}, \qquad \star \in \{N, W\},\ A \ge 1, \tag{6.5}$$

for every deterministic nonempty integer interval I. □

7 The endpoint modulus and graph limits

7.1 Uniform estimates on small clock intervals

For $R > 0$, $0 < \delta < 1$, and $n \ge 1$, let $\Xi_{n,R}(\delta)$ be $n^{-\sigma}$ times the largest of

$$C_T(I), \quad T \in \mathcal{T}, \qquad \text{and} \qquad A^\star(I), \quad \star \in \{N, W\},$$

where in each case I is an integer interval in the relevant clock, contained in $[-Rn-10, Rn+10]$, with $|I| \le \delta n + 10$.

Lemma 7.1 (Small-interval costs)**.** *For every $R, \varepsilon > 0$,*

$$\lim_{\delta \downarrow 0} \limsup_{n \to \infty} \mathbb{P}[\Xi_{n,R}(\delta) > \varepsilon] = 0. \tag{7.1}$$

Proof. Let $h = \lceil \delta n \rceil + 10$ and use deterministic intervals

$$J_q = [qh, (q+2)h] \cap \mathbb{Z}.$$

Every interval in the definition of $\Xi_{n,R}(\delta)$ is contained in one such J_q, and at most $C_R(1+\delta^{-1})$ intervals are needed. For fixed δ and all sufficiently large n, $|J_q| \le C\delta n$. The costs are monotone under interval inclusion. Therefore Propositions 4.2 and (6.5), followed by a union bound, give

$$\mathbb{P}[\Xi_{n,R}(\delta) > \varepsilon] \le C_R(1+\delta^{-1}) \left[\exp\{-c\sqrt{\varepsilon}\,\delta^{-3/16}\} + \exp\{-c\varepsilon\,\delta^{-3/8}\} \right].$$

After changing constants. The right side tends to zero as $\delta \downarrow 0$. No independence is used. □

7.2 Endpoint comparison, including the order boundary

Put $K_R = [-R, R]^4$ and

$$\omega_{n,R}(\delta) = \sup\{|D_n(w) - D_n(w')| : w, w' \in W_n \cap K_R,\ \|w - w'\|_\infty \le \delta\}, \tag{7.2}$$

with an empty supremum equal to zero.

Proposition 7.2 (Stochastic equicontinuity)**.** *For every $R, \varepsilon > 0$,*

$$\lim_{\delta \downarrow 0} \limsup_{n \to \infty} \mathbb{P}[\omega_{n,R}(\delta) > \varepsilon] = 0. \tag{7.3}$$

Proof. Work first with raw black points. Let (a, b) and (a', b') be comparable pairs whose four normalized Peano coordinates differ by at most δ, equivalently whose raw coordinates differ by at most δn.

If $a' \preceq b$, then moving the source and then the target is legitimate. The comparisons (2.12)–(2.13), with (4.3) and (4.8), give

$$|d(a, b) - d(a', b')| \le K(a, a') + K(b, b'). \tag{7.4}$$

Every interval on the right has cardinality at most $\delta n + 1$.

Suppose instead that $a' \not\preceq b$. By (1.3), there is a coordinate $q \in \{N, W\}$ such that the q-label of a' is strictly larger than that of b. Since $a \preceq b$ and $a' \preceq b'$, and the two base pairs are δn-close coordinatewise,

$$0 \le q(b) - q(a) \le \delta n, \qquad 0 \le q(b') - q(a') \le \delta n. \tag{7.5}$$

Thus $d(a, b)$ and $d(a', b')$ are each bounded by the anchored diameter of an interval of cardinality at most $\delta n + 1$ in the same clock q. No cross-pair comparability is needed in this case.

For base points in K_R, all intervals just described lie in the window used for $\Xi_{n,R+1}(\delta)$ for all sufficiently large n. Hence, after a universal enlargement of δ absorbing the $O(1)$ endpoints,

$$\omega_{n,R}(\delta) \le C \frac{n^\sigma}{m_n}\, \Xi_{n,R+1}(C\delta) + \frac{C}{m_n}.$$

By (1.8), n^σ / m_n is uniformly bounded and $m_n \to \infty$. Lemma 7.1 proves (7.3). Since $d_M^\bullet = d + 1$ on comparable black points, changing back to the original edge-count convention changes no oscillation. □

Proposition 7.3 (Local height bounds)**.** *For each $R > 0$, the random variables*

$$H_{n,R} = \sup\{D_n(w) : w \in W_n \cap K_R\} \tag{7.6}$$

form a tight family.

Proof. For comparable black points represented in K_R, both their N and W clock intervals are contained in $[-\lceil Rn\rceil, \lceil Rn\rceil]$. By (4.9),

$$m_n H_{n,R} \le C_{T_R^+}([-\lceil Rn\rceil, \lceil Rn\rceil]) + C_{T_R^-}([-\lceil Rn\rceil, \lceil Rn\rceil]) + 1.$$

Use Proposition 4.2 and the lower bound in (1.8). □

7.3 Identification of every limiting fiber

Proof of Theorem 1.1. Fix a subsequence along which (W_n, Γ_n) converges weakly. By Proposition 7.3, the countable vector $(H_{n,R})_{R\in\mathbb{N}}$ is tight: for each $\eta > 0$, choose K_R so that $\sup_n \mathbb{P}[H_{n,R} > K_R] \le \eta 2^{-R}$ and take the product compact set $\prod_R [0, K_R]$. Hence we may adjoin this vector and pass to a further subsequence along which the enlarged vector converges weakly, with every limiting height finite. The Fell hyperspace of a locally compact second-countable space is Polish, and so is its countable product with $[0,\infty)^{\mathbb{N}}$. By Skorokhod representation we may work on a probability space on which, almost surely,

$$W_n \to W, \qquad \Gamma_n \to \Gamma, \qquad H_{n,R} \to H_R < \infty \quad (R \in \mathbb{N}). \tag{7.7}$$

The marginal laws of the prelimit variables are unchanged, so all estimates above remain valid.

We first show pathwise that the projection of Γ onto its first four coordinates equals W. If $(w,t) \in \Gamma$, Fell convergence gives graph points $(w_n, D_n(w_n)) \to (w,t)$, so $w_n \in W_n$ and therefore $w \in W$. Conversely, fix $w \in W$. Fell convergence of W_n supplies $w_n \in W_n$ with $w_n \to w$. Choose R so that $w_n \in K_R$ eventually. By (7.7), $D_n(w_n) \le H_{n,R}$ is bounded, so some subsequence converges to a finite t. Then $(w,t) \in \Gamma$. The same argument, using $H_{n,R+1}$, shows that the heights of Γ above each compact base box are bounded.

It remains to exclude multiple heights over one base point. Fix integers $R, A \ge 1$ and $\varepsilon > 0$. Let $E_{R,A,\varepsilon}$ be the event that Γ contains (w,s) and (w,t) with

$$w \in K_R, \qquad s,t \in [0,A], \qquad |s-t| \ge \varepsilon.$$

This event is Borel. Indeed, after restricting the two points to the compact set $K_R \times [0,A]$, the incidence relation $\{(F,x) : x \in F\}$ is closed for Fell convergence; imposing equality of the base coordinates and height separation at least ε gives a closed subset of the product with a compact point space, and its projection to the hyperspace is closed.

On $E_{R,A,\varepsilon}$, Fell convergence supplies graph points

$$(w_n, D_n(w_n)) \to (w,s), \qquad (w_n', D_n(w_n')) \to (w,t).$$

For every fixed $\delta > 0$, eventually $w_n, w_n' \in K_{R+1}$, $\|w_n - w_n'\|_\infty < \delta$, and $|D_n(w_n) - D_n(w_n')| > \varepsilon/2$. Therefore, pathwise on the Skorokhod space,

$$E_{R,A,\varepsilon} \subset \liminf_{n\to\infty} \{\omega_{n,R+1}(\delta) > \varepsilon/2\}. \tag{7.8}$$

Taking probabilities, then using Fatou in the form $\mathbb{P}(\liminf E_n) \le \liminf \mathbb{P}(E_n)$ for the inclusion above, gives

$$\mathbb{P}(E_{R,A,\varepsilon}) \le \liminf_{n\to\infty} \mathbb{P}[\omega_{n,R+1}(\delta) > \varepsilon/2].$$

Let $\delta \downarrow 0$ and apply Proposition 7.2. A countable union over $R, A \in \mathbb{N}$ and positive rational ε shows that every fiber of Γ is a singleton almost surely.

Write $\Gamma = \mathrm{Graph}(W, D)$. The projection argument already gives local boundedness of D. If $w_k \to w$ in W, the values $D(w_k)$ are bounded in a compact base box. Every convergent subsequence has a limit t with $(w, t) \in \Gamma$ by closedness of Γ, and the singleton-fiber property forces $t = D(w)$. Hence the whole sequence satisfies $D(w_k) \to D(w)$, proving continuity.

The original weak subsequential limit has the same (W, Γ) marginal as the further subsequence used for the Skorokhod argument, so the conclusion applies to the original limit. Finally, the $+1$ shift between $d_M^\bullet$ and d tends to zero after division by m_n by (1.8). This proves the theorem for the graphs defined in (1.6). □

A Five-pass adversarial audit record

The following records five logically separate checks carried out after the revisions above. They are mathematical self-audits, not external refereeing and not proof-assistant verification.

A.1 Pass I: deterministic KMSW and planar geometry

The signed endpoint tests were reconstructed directly from the two frontier stacks. The two-buffer argument was checked using only anchored shortcuts, so no missing boundary edge can create an unaccounted shortcut. The four extremal-tree intersection statements were checked in an upward embedding, including the case of repeated vertices and pinches.

Two previous points of compression were expanded. First, the infinite-volume two-order characterization now includes an explicit local-stabilization argument for finite segments of both primal Peano orders before invoking the finite Baxter product-order theorem. Second, Lemma 3.2 isolates the exact cap crosscut needed in Section 3; the extension from x_0 to every x_i is reduced to the published x_0 separation argument by concatenating the right-boundary path. Third, Lemma 5.4 makes the two record-edge barriers in Section 5 quantitative rather than leaving their enclosure property implicit.

A.2 Pass II: external inputs and symmetry conventions

The boundary input is exactly [2, Lemma 7.8]: simultaneous short paths from every lower-boundary vertex to the upper boundary. No assertion about a short initial segment of an infinite geodesic is used. The cap law and its reversal/reflection symmetry are taken from [2, Lemmas 7.9 and 7.13], while (3.4) uses only their stopped-time coupling.

For transverse labels, $M_{0,t}^N$ and $M_{0,t}^S$ are the regions of [5, Section 5.1]. The north conditional law is exactly [5, Lemma 5.2]. The south law is not quoted with the wrong sigma-field: it is derived by applying that lemma to the explicitly reversed/reflected encoding $\check{Z}_t$. The primal west-going walk, and only that walk, is assigned the ordinary i.i.d. three-step law, in accordance with [5, Remark 1.8]; equality of primal and dual west-going edge orders uses [5, Lemma 2.5].

A.3 Pass III: probability and certificate arguments

The cell recursion and boundary recursion use Reimer's inequality only after exhibiting disjoint coordinate certificates that remain valid under arbitrary resampling of the complementary coordinates. The boundary coordinates are whole descending-excursion words, so random excursion lengths do not spoil the product structure. The passage from a countable excursion alphabet to finite alphabets is by total-variation approximation.

In the stopped-pocket argument, the future estimate uses the ordinary stopping-time property, whereas the past estimate conditions on $\mathcal{G}_0$, with respect to which both T_n and the good event are measurable. No claim that T_n is a stopping time for a reversed filtration is made. Every combination of random branch length and boundary cost is by a union bound, not by an independence assertion.

A.4 Pass IV: exponents, normalization, and dependency order

The exponent arithmetic was recomputed independently. A boundary index range of order r is reached after ordinary Peano time of order r^2, and the KMSW-cell SDP scale $(r^2)^{3/8}$ is $r^{3/4}$. In Proposition 4.2, choosing branch length $r \asymp A^{2/3}m^{1/2}$ makes

$$r^2/m \asymp A^{4/3}, \qquad \frac{Am^{3/8}}{r^{3/4}} \asymp A^{1/2},$$

which gives the stated $e^{-c\sqrt{A}}$ tail. For intervals of size δn, the two tails become $\exp\{-c\sqrt{\varepsilon}\,\delta^{-3/16}\}$ and $\exp\{-c\varepsilon\delta^{-3/8}\}$, which dominate the $O(\delta^{-1})$ covering number.

The logical dependency is acyclic: the stated Borga–Gwynne boundary crossing estimate gives the half-map boundary upper quantile; Reimer gives the boundary exponential tail; this gives tree costs. Independently, the GHS pocket law gives transverse compactness. Tree costs plus transverse compactness give a cell upper quantile. Only then is the cell Reimer recursion invoked for the sharp exponential cell tail. The lower median bound is the third item of [2, Theorem 1.13]. Thus the proof of $m_n \asymp n^{3/8}$ does not assume that comparison.

A.5 Pass V: Fell topology and limiting quantifiers

The limiting argument was checked without inserting an almost-sure equicontinuity statement that is not proved. Local height tightness is adjoined before Skorokhod representation, yielding pathwise finite local height bounds. Exact projection onto the limiting base is then proved directly from Fell convergence. Multiple fibers are excluded using the fixed-δ event inclusion (7.8) and stochastic equicontinuity; the double-fiber event is explicitly Borel. Finally, closedness of the singleton graph together with local boundedness implies sequential continuity on the closed domain W.

No uniqueness of the subsequential law, no metric axioms for the limiting function, and no measurability of a limiting metric from a continuum random surface are asserted.

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